

# AVANZA

## Foreign Degree Validation Platform

---

### Product Requirements Document

|                  |                             |
|------------------|-----------------------------|
| Document Version | 1.0                         |
| Date             | November 2024               |
| Author           | Avanza Product Team         |
| Status           | Draft — For Review          |
| Confidentiality  | Internal + Development Team |

# Table of Contents

|     |                             |
|-----|-----------------------------|
| 1   | Executive Summary           |
| 2   | Product Vision & Goals      |
| 3   | Target Users & Personas     |
| 4   | Product Scope               |
| 5   | Functional Requirements     |
| 5.1 | Public Website              |
| 5.2 | Client Portal               |
| 5.3 | Admin Dashboard             |
| 6   | Non-Functional Requirements |
| 7   | User Stories                |
| 8   | Information Architecture    |
| 9   | Data Model                  |
| 10  | API Specifications          |
| 11  | Third-Party Integrations    |

# 1. Executive Summary

## 1.1 Problem Statement

Millions of foreign nationals living in Italy hold professional qualifications and degrees from their home countries that are not recognized in Italy. The process to validate these credentials is:

- **Complex:** Multiple pathways, different authorities, varying requirements by country and profession
- **Bureaucratic:** Extensive paperwork, long timelines, confusing procedures
- **Fragmented:** No single point of service; requires interaction with multiple institutions
- **Language-dependent:** Most guidance is only available in Italian
- **Error-prone:** Small mistakes lead to rejections and months of delay

Currently, these individuals must either:

- Navigate the process alone (high failure rate, 6–18 months)
- Use expensive immigration lawyers (€2,000–5,000+)
- Visit overwhelmed patronati offices (free but extremely slow and generic)

## 1.2 Solution

**Avanza** is a digital platform that guides foreign nationals through the entire degree/credential validation process in Italy, from initial assessment to successful recognition and professional registration.

The platform provides:

- Intelligent assessment to identify the correct recognition pathway
- Document management and preparation
- Application submission and tracking
- Real-time communication with expert team
- End-to-end case management

## 1.3 Business Model

- **Service-based revenue:** Consulting fees and service packages (€99–€2,500)
- **Tiered packages:** Starter, Standard, Premium
- **Add-on services:** Translations, business registration, expedited processing
- **B2B contracts:** Companies needing to validate employee credentials

## 1.4 Key Differentiators

| # | Differentiator | Description                                                                  |
|---|----------------|------------------------------------------------------------------------------|
| 1 | Specialization | 100% focused on credential recognition (unlike generic immigration services) |
| 2 | Digital-first  | Fully online process with client portal (unlike office-based patronati)      |
| 3 | Multilingual   | 10+ languages supported (unlike Italian-only resources)                      |
| 4 | Affordable     | €400–1,500 average (between free/slow and expensive/lawyer)                  |
| 5 | Transparent    | Clear pricing, timelines, and real-time progress tracking                    |
| 6 | Intelligent    | Smart assessment quiz, personalized recommendations, automation              |

## 2. Product Vision & Goals

### 2.1 Product Vision

*"Make credential recognition in Italy accessible, transparent, and stress-free for every qualified professional, regardless of where they studied."*

### 2.2 Mission Statement

*"To become Italy's leading platform for foreign credential validation by providing expert guidance, digital tools, and human support that transforms a confusing bureaucratic process into a clear, manageable journey."*

### 2.3 Business Goals

| Goal                             | Target         | Timeline  |
|----------------------------------|----------------|-----------|
| Acquire first 50 paying clients  | 50 clients     | Month 1–3 |
| Achieve €10,000 monthly revenue  | €10K MRR       | Month 4–6 |
| Reach 200+ active cases          | 200 cases      | Month 6–9 |
| Achieve 90%+ client satisfaction | NPS > 50       | Month 6+  |
| Achieve 85%+ success rate        | 85% approvals  | Ongoing   |
| Reach €50,000 monthly revenue    | €50K MRR       | Month 12  |
| Serve clients from 20+ countries | 20 countries   | Month 12  |
| Support 30+ profession types     | 30 professions | Month 12  |

### 2.4 Product Goals

| Goal                | Metric                           | Target                          |
|---------------------|----------------------------------|---------------------------------|
| Easy onboarding     | Quiz completion rate             | >70%                            |
| Quick time-to-value | Time from signup to first action | <10 minutes                     |
| High engagement     | Portal login frequency           | 2+ times/week                   |
| Low support burden  | Self-service resolution rate     | >60%                            |
| Fast response time  | Team first reply time            | <4 hours (business)             |
| Smooth payments     | Payment completion rate          | >90%                            |
| Strong SEO presence | Organic traffic                  | 10,000+ visits/month by Month 6 |
| High conversion     | Quiz → paid client               | >5%                             |

### 3. Target Users & Personas

#### 3.1 Primary Persona: "Alexandru" — EU Worker

|                         |                                                                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                    | Alexandru Popescu                                                                                                                              |
| Age                     | 34                                                                                                                                             |
| Nationality             | Romanian                                                                                                                                       |
| Lives in                | Milan, Italy (since 2021)                                                                                                                      |
| Profession              | Electrician (diploma from Romania, 2012)                                                                                                       |
| Language                | Romanian (native), Italian (intermediate), English (basic)                                                                                     |
| Digital literacy        | Medium (uses smartphone daily, comfortable with WhatsApp)                                                                                      |
| Goal                    | Open his own electrical installation company in Italy                                                                                          |
| Pain points             | Doesn't understand Italian bureaucracy; tried Camera di Commercio but confused; can't afford a lawyer (€3,000+); patronato gave generic advice |
| Budget                  | €500–1,000                                                                                                                                     |
| Preferred communication | WhatsApp, Italian or Romanian                                                                                                                  |

#### 3.2 Secondary Persona: "Olena" — Non-EU Professional

|                         |                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Name                    | Olena Kovalenko                                                                                                                              |
| Age                     | 29                                                                                                                                           |
| Nationality             | Ukrainian                                                                                                                                    |
| Lives in                | Rome, Italy (since 2022, refugee status)                                                                                                     |
| Profession              | Nurse (Bachelor's degree from Kyiv Medical University)                                                                                       |
| Language                | Ukrainian (native), Russian, English (good), Italian (learning)                                                                              |
| Digital literacy        | High (uses apps, email, comfortable online)                                                                                                  |
| Goal                    | Work as a registered nurse in Italian hospitals                                                                                              |
| Pain points             | More complex for non-EU; needs Ministry of Health recognition; may need language/professional exam; emotionally stressed; needs to work ASAP |
| Budget                  | €800–1,500                                                                                                                                   |
| Preferred communication | Email, Telegram, English or Ukrainian                                                                                                        |

#### 3.3 Tertiary Persona: "Ahmed" — Skilled Tradesperson

|             |                                                            |
|-------------|------------------------------------------------------------|
| Name        | Ahmed Benali                                               |
| Age         | 41                                                         |
| Nationality | Moroccan                                                   |
| Lives in    | Turin, Italy (since 2015, permanent residence)             |
| Profession  | Mechanical Engineer (degree from University of Casablanca) |
| Language    | Arabic (native), French, Italian (good)                    |

|                         |                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Digital literacy        | Medium                                                                                                                   |
| Goal                    | Get engineering degree recognized to advance career / better salary                                                      |
| Pain points             | Working below qualification; non-EU complex process (needs Dichiarazione di Valore); previous attempt failed; frustrated |
| Budget                  | €1,000–2,000                                                                                                             |
| Preferred communication | WhatsApp, Italian or French                                                                                              |

### 3.4 B2B Persona: "Marco" — Italian Employer

|                         |                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|
| Name                    | Marco Rossi                                                                                                         |
| Age                     | 48                                                                                                                  |
| Role                    | HR Manager at a construction company (150 employees)                                                                |
| Goal                    | Verify and validate credentials of foreign employees                                                                |
| Pain points             | Legal liability; doesn't know how to verify foreign credentials; needs bulk processing; time-consuming individually |
| Budget                  | €500–2,000 per employee (company pays)                                                                              |
| Preferred communication | Email, Italian                                                                                                      |

### 3.5 User Segments Summary

| Segment                                  | % of Market | Avg. Budget  | Complexity  | Priority |
|------------------------------------------|-------------|--------------|-------------|----------|
| EU workers (trades)                      | 35%         | €500–800     | Medium      | High     |
| Non-EU professionals (healthcare)        | 25%         | €1,000–2,000 | High        | High     |
| EU professionals (engineering, teaching) | 20%         | €800–1,500   | Medium-High | Medium   |
| International students                   | 10%         | €300–600     | Low-Medium  | Medium   |
| B2B (companies)                          | 5%          | €2,000–5,000 | Varies      | Future   |
| Non-EU trades                            | 5%          | €600–1,200   | High        | Medium   |

## 4. Product Scope

### 4.1 In Scope (Phase 1 — MVP)

| Category          | Included                                                                |
|-------------------|-------------------------------------------------------------------------|
| User types        | Clients (B2C), Admin team                                               |
| Countries         | Romania, Albania, Ukraine, Morocco, India (top 5)                       |
| Professions       | Electrician, Nurse, Engineer, Teacher, Construction worker (top 5)      |
| Recognition types | EU Professional Recognition, Dichiarazione di Valore, CIMEA attestation |
| Languages         | Italian, English, Romanian                                              |
| Platform          | Web application (responsive, mobile-friendly)                           |
| Payments          | Credit card (Stripe), PayPal, bank transfer                             |
| Communication     | In-app messaging, email, WhatsApp                                       |

### 4.2 In Scope (Phase 2 — Growth)

| Category          | Included                                                          |
|-------------------|-------------------------------------------------------------------|
| Countries         | Expand to 15+ countries                                           |
| Professions       | Expand to 20+ professions                                         |
| Languages         | Add Albanian, Arabic, Ukrainian, French, Spanish                  |
| Features          | Cost calculator, automation, advanced analytics, referral program |
| Recognition types | Add Equipollenza, Professional Order registration                 |

### 4.3 In Scope (Phase 3 — Scale)

| Category   | Included                                        |
|------------|-------------------------------------------------|
| User types | Add B2B portal                                  |
| Countries  | All countries                                   |
| Languages  | 10+ languages                                   |
| Features   | AI document checker, mobile app, community, API |
| Platform   | PWA + potential native app                      |

### 4.4 Out of Scope

| Item                                      | Reason                                        |
|-------------------------------------------|-----------------------------------------------|
| Legal representation in court             | Requires licensed lawyer                      |
| Sworn translation services (in-house)     | Requires certified translator                 |
| Issuing official recognitions             | Only government authorities can do this       |
| Immigration/visa services                 | Different service; potential future expansion |
| Credential validation for other countries | Focus on Italy first                          |
| Academic tutoring/exam prep               | Not core service                              |

| Item                   | Reason                 |
|------------------------|------------------------|
| Physical office visits | Digital-first approach |



## 5. Functional Requirements

### 5.1 Public Website

#### FR-PW-001: Homepage

| ID           | Requirement                                                       | Priority | Acceptance Criteria                                         |
|--------------|-------------------------------------------------------------------|----------|-------------------------------------------------------------|
| FR-PW-001.1  | Display hero section with primary value proposition               | Must     | Hero visible above fold on all devices                      |
| FR-PW-001.2  | Show dynamic statistics (clients served, success rate, countries) | Must     | Numbers update from database, cached for performance        |
| FR-PW-001.3  | Display 'How it Works' section with 4 steps                       | Must     | Steps are interactive, link to relevant pages               |
| FR-PW-001.4  | Show service packages with pricing                                | Must     | Pricing pulled from admin-configurable settings             |
| FR-PW-001.5  | Display testimonial carousel                                      | Must     | Minimum 3 testimonials visible, auto-rotate every 5 seconds |
| FR-PW-001.6  | Show country selection (interactive)                              | Should   | Clickable countries with flag icons, link to country pages  |
| FR-PW-001.7  | FAQ section with accordion                                        | Must     | Schema markup for SEO, expandable/collapsible               |
| FR-PW-001.8  | CTA buttons visible without scrolling                             | Must     | Primary CTA above fold, secondary CTAs throughout           |
| FR-PW-001.9  | Language selector in header                                       | Must     | Change language without page reload (SPA) or with redirect  |
| FR-PW-001.10 | Floating WhatsApp button                                          | Must     | Fixed position, visible on all pages, opens WhatsApp chat   |

#### FR-PW-002: Assessment Quiz

| ID           | Requirement                                        | Priority | Acceptance Criteria                                                |
|--------------|----------------------------------------------------|----------|--------------------------------------------------------------------|
| FR-PW-002.1  | Multi-step form with 8 screens                     | Must     | Each screen loads without full page refresh                        |
| FR-PW-002.2  | Progress indicator showing current step            | Must     | Visual bar or numbered steps                                       |
| FR-PW-002.3  | Conditional logic between questions                | Must     | Next question adapts based on previous answer                      |
| FR-PW-002.4  | Country selector with search and flags             | Must     | Searchable dropdown, shows flag icon, supports 200+ countries      |
| FR-PW-002.5  | Profession selector with categories                | Must     | Grouped by category, searchable, 'Other' option with free text     |
| FR-PW-002.6  | Document checklist with checkboxes                 | Must     | Pre-populated based on country+profession combo                    |
| FR-PW-002.7  | Email capture on final step (required for results) | Must     | Email validation, GDPR consent checkbox                            |
| FR-PW-002.8  | Results page with personalized recommendation      | Must     | Shows: pathway, documents needed, timeline, cost estimate, package |
| FR-PW-002.9  | Save partial progress (browser session)            | Should   | If user closes and reopens within 24h, resume where left off       |
| FR-PW-002.10 | Results sent via email                             | Must     | Triggered immediately after completion                             |
| FR-PW-002.11 | CRM lead creation on completion                    | Must     | All quiz data stored as lead properties                            |

| ID           | Requirement                          | Priority | Acceptance Criteria                                                   |
|--------------|--------------------------------------|----------|-----------------------------------------------------------------------|
| FR-PW-002.12 | Analytics tracking per step          | Must     | Track: completion rate, drop-off point, time per step                 |
| FR-PW-002.13 | Available in all supported languages | Must     | All quiz text translatable                                            |
| FR-PW-002.14 | Mobile-optimized                     | Must     | Tested on iOS Safari + Android Chrome                                 |
| FR-PW-002.15 | Results logic configurable by admin  | Should   | Admin can modify which country+profession combo leads to which result |

## Quiz Logic Engine — Output Specification

### INPUT:

- Country of qualification
- Qualification type (vocational/bachelor/master/PhD)
- Profession
- Goal (employment/self-employment/study)
- Residence status (in Italy/abroad/planning)
- Years of experience
- Documents already held

### OUTPUT (determined by logic rules):

- Recognition pathway type:
  - EU Automatic Recognition (Annex IV/V of Directive 2005/36/EC)
  - EU General System Recognition (with possible compensatory measures)
  - Dichiarazione di Valore + Ministry recognition
  - CIMEA credential evaluation
  - University Equipollenza
  - Professional Order registration
- Competent authority (MIMIT, Camera di Commercio, Ministry of Health, etc.)
- Required documents list (specific to their situation)
- Estimated timeline
- Estimated cost range
- Recommended service package
- Potential obstacles/notes

## FR-PW-003: Service Pages

| ID          | Requirement                                    | Priority | Acceptance Criteria                                |
|-------------|------------------------------------------------|----------|----------------------------------------------------|
| FR-PW-003.1 | Service overview page with all services        | Must     | Card grid layout, filterable by category           |
| FR-PW-003.2 | Individual service detail page (template)      | Must     | Consistent structure across all services           |
| FR-PW-003.3 | Package comparison table                       | Must     | Side-by-side feature comparison, highlight popular |
| FR-PW-003.4 | 'Add to cart' / 'Select Package' functionality | Must     | Links to checkout/booking flow                     |
| FR-PW-003.5 | Pricing configurable by admin                  | Must     | No code changes needed to update prices            |

## FR-PW-004: Country Pages

| ID          | Requirement                                      | Priority | Acceptance Criteria                           |
|-------------|--------------------------------------------------|----------|-----------------------------------------------|
| FR-PW-004.1 | Template-based country page                      | Must     | Same structure, different content per country |
| FR-PW-004.2 | Country overview (EU/non-EU, special agreements) | Must     | Accurate legal framework info                 |
| FR-PW-004.3 | Common professions from this country (linked)    | Must     | Click → goes to country+profession combo page |
| FR-PW-004.4 | Required documents specific to this country      | Must     | Names in original language + Italian          |

| ID          | Requirement                                      | Priority | Acceptance Criteria           |
|-------------|--------------------------------------------------|----------|-------------------------------|
| FR-PW-004.5 | Institutions in home country (where to get docs) | Must     | Names, websites, contact info |

### FR-PW-005: Profession Pages

| ID          | Requirement                            | Priority | Acceptance Criteria                              |
|-------------|----------------------------------------|----------|--------------------------------------------------|
| FR-PW-005.1 | Template-based profession page         | Must     | Same structure, different content per profession |
| FR-PW-005.2 | Italian regulatory framework explained | Must     | Which law, which authority, what requirements    |
| FR-PW-005.3 | Recognition requirements breakdown     | Must     | Qualification + experience requirements          |
| FR-PW-005.4 | Country links                          | Must     | Click → goes to country+profession combo page    |
| FR-PW-005.5 | Profession-specific timeline and costs | Must     | Realistic estimates                              |

### FR-PW-006: Combination Pages (Country + Profession)

| ID          | Requirement                                        | Priority | Acceptance Criteria                            |
|-------------|----------------------------------------------------|----------|------------------------------------------------|
| FR-PW-006.1 | Detailed step-by-step guide                        | Must     | Complete roadmap for this specific combination |
| FR-PW-006.2 | Generated from country + profession data           | Must     | Auto-populated but admin-overridable           |
| FR-PW-006.3 | Specific documents with names in original language | Must     | e.g., 'Diplom de calificare profesional'       |
| FR-PW-006.4 | Specific institutions to contact                   | Must     | With links and instructions                    |
| FR-PW-006.5 | HowTo schema markup                                | Must     | For Google rich snippets                       |
| FR-PW-006.6 | CTA to start service                               | Must     | 'Let us handle this for €X'                    |

### FR-PW-008: Booking System (Public)

| ID           | Requirement                        | Priority | Acceptance Criteria                                               |
|--------------|------------------------------------|----------|-------------------------------------------------------------------|
| FR-PW-008.1  | Appointment type selection         | Must     | List of available consultation types with descriptions and prices |
| FR-PW-008.2  | Calendar view with available slots | Must     | Shows only available dates/times                                  |
| FR-PW-008.3  | Timezone detection and selection   | Must     | Auto-detect, allow manual override                                |
| FR-PW-008.4  | Client information form            | Must     | Name, email, phone, brief case description                        |
| FR-PW-008.5  | Payment for paid consultations     | Must     | Integrated payment before confirmation                            |
| FR-PW-008.7  | Confirmation page and email        | Must     | Details + calendar invite (.ics) + video link                     |
| FR-PW-008.8  | Automated reminders                | Must     | 24h before + 1h before (email + SMS/WhatsApp)                     |
| FR-PW-008.9  | Reschedule/cancel functionality    | Must     | Client can reschedule within policy                               |
| FR-PW-008.12 | Video meeting link auto-generation | Must     | Google Meet or Zoom link in confirmation                          |

### FR-PW-010: Multi-Language System

| ID          | Requirement                               | Priority | Acceptance Criteria                      |
|-------------|-------------------------------------------|----------|------------------------------------------|
| FR-PW-010.1 | Language selector in header (all pages)   | Must     | Flag + language name, dropdown           |
| FR-PW-010.2 | URL-based language routing                | Must     | /it/, /en/, /ro/ prefix or subdomain     |
| FR-PW-010.3 | Browser language detection on first visit | Must     | Suggest switch if browser lang ≠ current |

| ID           | Requirement                            | Priority         | Acceptance Criteria                             |
|--------------|----------------------------------------|------------------|-------------------------------------------------|
| FR-PW-010.4  | Language preference persistence        | Must             | Save in cookie/localStorage/user profile        |
| FR-PW-010.5  | All UI strings translatable            | Must             | No hardcoded strings                            |
| FR-PW-010.6  | CMS content translatable per page      | Must             | Each page can have variants per language        |
| FR-PW-010.7  | RTL support for Arabic/Urdu            | Should (Phase 2) | Full RTL layout                                 |
| FR-PW-010.8  | Hreflang tags                          | Must             | Correct alternate language links for SEO        |
| FR-PW-010.9  | Fallback language                      | Must             | If content not translated, show Italian/English |
| FR-PW-010.11 | Admin translation management interface | Must             | See what's translated, what's missing           |

**Phase 1 Languages:** Italian, English, Romanian

**Phase 2 Languages:** + Albanian, Arabic, Ukrainian

**Phase 3 Languages:** + French, Spanish, Russian, Hindi, Bengali, Urdu

## 5.2 Client Portal

### FR-CP-001: Authentication

| ID           | Requirement                             | Priority | Acceptance Criteria                                  |
|--------------|-----------------------------------------|----------|------------------------------------------------------|
| FR-CP-001.1  | Email + password registration           | Must     | Email validation, password strength meter            |
| FR-CP-001.2  | Email verification                      | Must     | Confirmation link sent, must verify to access portal |
| FR-CP-001.3  | Social login (Google)                   | Must     | One-click sign in with Google                        |
| FR-CP-001.5  | Login with email + password             | Must     | With 'Remember me' option                            |
| FR-CP-001.6  | Forgot password flow                    | Must     | Email reset link, expires in 1 hour                  |
| FR-CP-001.7  | Password requirements                   | Must     | Min 8 chars, 1 uppercase, 1 number                   |
| FR-CP-001.8  | Account lockout after 5 failed attempts | Must     | Locked 30 min, notification sent                     |
| FR-CP-001.9  | Two-factor authentication (optional)    | Should   | TOTP or SMS-based                                    |
| FR-CP-001.10 | Session timeout after 30 min inactivity | Must     | Redirect to login                                    |
| FR-CP-001.12 | New device login alert                  | Must     | Email notification on new device/location            |

### FR-CP-002: User Profile

| ID           | Requirement                     | Priority | Acceptance Criteria                                             |
|--------------|---------------------------------|----------|-----------------------------------------------------------------|
| FR-CP-002.1  | Personal information fields     | Must     | Name, email, phone, DOB, nationality, residence, codice fiscale |
| FR-CP-002.2  | Professional information fields | Must     | Country of qualification, type, profession, experience years    |
| FR-CP-002.4  | Profile completeness indicator  | Must     | Percentage bar, prompts to complete                             |
| FR-CP-002.5  | Edit and save changes           | Must     | Inline editing, save button, validation                         |
| FR-CP-002.6  | Change password                 | Must     | Current password required                                       |
| FR-CP-002.8  | Communication preferences       | Must     | Choose channels: email, WhatsApp, SMS, push                     |
| FR-CP-002.10 | Delete account                  | Must     | GDPR compliance, confirmation step, data deletion               |
| FR-CP-002.11 | Export my data                  | Must     | GDPR compliance, downloadable ZIP/JSON                          |

### FR-CP-003: Client Dashboard

| ID           | Requirement                     | Priority | Acceptance Criteria                          |
|--------------|---------------------------------|----------|----------------------------------------------|
| FR-CP-003.1  | Personalized welcome message    | Must     | 'Welcome back, [Name]!'                      |
| FR-CP-003.2  | Active case summary card(s)     | Must     | Case ID, title, progress %, last update time |
| FR-CP-003.3  | Visual progress bar per case    | Must     | Percentage + current step name               |
| FR-CP-003.4  | Action required alerts          | Must     | Highlighted, with direct action link         |
| FR-CP-003.5  | Recent messages preview         | Must     | Latest 2–3 messages, click to see full       |
| FR-CP-003.6  | Upcoming appointments/deadlines | Must     | Next 3 items, chronological                  |
| FR-CP-003.7  | Document status summary         | Must     | X uploaded, Y verified, Z pending            |
| FR-CP-003.8  | Quick action buttons            | Must     | Upload, Message, Book, Pay                   |
| FR-CP-003.10 | Real-time updates               | Must     | Dashboard refreshes without manual reload    |

### FR-CP-004: Case View

| ID           | Requirement                                            | Priority | Acceptance Criteria                                        |
|--------------|--------------------------------------------------------|----------|------------------------------------------------------------|
| FR-CP-004.1  | Case overview (title, ID, dates, assigned team member) | Must     | All key info visible at top                                |
| FR-CP-004.2  | Visual timeline with steps                             | Must     | Vertical timeline, status per step (done/current/upcoming) |
| FR-CP-004.3  | Current step highlighted                               | Must     | Clearly distinguishable from completed/future steps        |
| FR-CP-004.5  | Activity log (chronological)                           | Must     | All events: uploads, messages, status changes, payments    |
| FR-CP-004.6  | Next steps / action items for client                   | Must     | Clear list of what client needs to do                      |
| FR-CP-004.7  | Deadline display with countdown                        | Must     | 'Apostille needed in 7 days' with visual urgency           |
| FR-CP-004.8  | Instructions/help per step                             | Must     | Expandable guide for current step                          |
| FR-CP-004.12 | Multiple cases support                                 | Must     | If client has >1 case, can switch between them             |

### FR-CP-005: Document Management

| ID           | Requirement                      | Priority | Acceptance Criteria                                           |
|--------------|----------------------------------|----------|---------------------------------------------------------------|
| FR-CP-005.1  | Document checklist per case      | Must     | Generated based on case type, admin-configurable              |
| FR-CP-005.2  | Upload via drag & drop           | Must     | Drag files to upload zone                                     |
| FR-CP-005.4  | Upload via camera (mobile)       | Must     | Access device camera, capture, upload                         |
| FR-CP-005.7  | File type validation             | Must     | Accept: PDF, JPG, PNG, TIFF, DOCX. Reject others with message |
| FR-CP-005.8  | File size limit (20MB per file)  | Must     | Error message if exceeded                                     |
| FR-CP-005.9  | Document status display          | Must     | Not uploaded / Uploaded / Under review / Verified / Rejected  |
| FR-CP-005.10 | Rejection reason display         | Must     | If rejected, show reason + fix instructions                   |
| FR-CP-005.11 | Re-upload (replace) capability   | Must     | Upload new version, old version archived                      |
| FR-CP-005.12 | In-browser preview (PDF, images) | Must     | View without downloading                                      |

| ID           | Requirement                | Priority | Acceptance Criteria                        |
|--------------|----------------------------|----------|--------------------------------------------|
| FR-CP-005.16 | Document deadline warnings | Must     | If document has a deadline, show countdown |
| FR-CP-005.19 | Storage encryption         | Must     | All files encrypted at rest                |
| FR-CP-005.20 | Virus scanning             | Must     | Scan every upload before storing           |

### FR-CP-006: Messaging System

| ID           | Requirement                              | Priority | Acceptance Criteria                                   |
|--------------|------------------------------------------|----------|-------------------------------------------------------|
| FR-CP-006.1  | Conversation view (chat-style)           | Must     | Messages displayed chronologically, sender identified |
| FR-CP-006.2  | Send text messages                       | Must     | Text input with send button                           |
| FR-CP-006.3  | Attach files in messages                 | Must     | Drag/drop or file picker within chat                  |
| FR-CP-006.6  | Real-time message delivery               | Must     | Messages appear immediately (WebSocket)               |
| FR-CP-006.7  | Read receipts                            | Should   | 'Seen' indicator                                      |
| FR-CP-006.8  | Typing indicator                         | Should   | 'Maria is typing...'                                  |
| FR-CP-006.9  | Message notifications                    | Must     | Badge count + email if not online                     |
| FR-CP-006.13 | Message per case (if multiple cases)     | Must     | Separate conversations per case                       |
| FR-CP-006.15 | Auto-notification if no reply in X hours | Must     | Escalation to team if message unanswered              |

### FR-CP-007: Payments (Client View)

| ID           | Requirement                              | Priority | Acceptance Criteria                      |
|--------------|------------------------------------------|----------|------------------------------------------|
| FR-CP-007.1  | View current balance/amount due          | Must     | Prominent display of outstanding payment |
| FR-CP-007.3  | Payment method: credit/debit card        | Must     | Via Stripe, support Visa/MC/Amex         |
| FR-CP-007.4  | Payment method: PayPal                   | Must     | PayPal checkout                          |
| FR-CP-007.5  | Payment method: bank transfer (bonifico) | Must     | Show bank details + reference code       |
| FR-CP-007.6  | Installment plans display                | Must     | Show payment schedule, paid/upcoming     |
| FR-CP-007.7  | Transaction history                      | Must     | Table of all past payments with status   |
| FR-CP-007.8  | Invoice download (PDF)                   | Must     | Italian-compliant invoice, downloadable  |
| FR-CP-007.10 | Payment receipt email                    | Must     | Auto-send on successful payment          |
| FR-CP-007.12 | Apply coupon/discount code               | Must     | Input field at checkout                  |

## 5.3 Admin Dashboard

### FR-AD-001: Admin Authentication & Access

| ID          | Requirement                                     | Priority | Acceptance Criteria                                         |
|-------------|-------------------------------------------------|----------|-------------------------------------------------------------|
| FR-AD-001.1 | Admin login (separate from client)              | Must     | Different login page or role-based redirect                 |
| FR-AD-001.2 | Two-factor authentication (mandatory for admin) | Must     | TOTP or hardware key                                        |
| FR-AD-001.3 | Role-based access control                       | Must     | Super Admin, Admin, Case Manager, Support, Content, Finance |
| FR-AD-001.4 | Permission matrix per role                      | Must     | Read/write/delete per module                                |

| ID          | Requirement        | Priority | Acceptance Criteria                             |
|-------------|--------------------|----------|-------------------------------------------------|
| FR-AD-001.5 | Activity audit log | Must     | Every admin action logged with user + timestamp |

## FR-AD-002: Case Management (Admin)

| ID           | Requirement                                            | Priority | Acceptance Criteria                                 |
|--------------|--------------------------------------------------------|----------|-----------------------------------------------------|
| FR-AD-002.1  | View all cases (list/table)                            | Must     | Sortable, filterable, searchable                    |
| FR-AD-002.2  | Kanban/pipeline view                                   | Must     | Drag & drop between stages                          |
| FR-AD-002.4  | Filter by: status, country, profession, assigned, date | Must     | Multi-filter combination                            |
| FR-AD-002.6  | Create new case manually                               | Must     | Form with all case fields                           |
| FR-AD-002.7  | Auto-create case on payment                            | Must     | When client pays, case created automatically        |
| FR-AD-002.8  | Case detail view                                       | Must     | All info: client, documents, timeline, notes, tasks |
| FR-AD-002.9  | Update case status                                     | Must     | Dropdown or button, triggers client notification    |
| FR-AD-002.10 | Assign case to team member                             | Must     | Dropdown selector, reassignment with notes          |
| FR-AD-002.11 | Internal notes (not visible to client)                 | Must     | Rich text, timestamped, per team member             |
| FR-AD-002.12 | Task creation per case                                 | Must     | Title, description, assignee, due date, status      |
| FR-AD-002.13 | Document verification (approve/reject)                 | Must     | Review uploaded docs, approve or reject with reason |
| FR-AD-002.16 | Case timeline/audit log                                | Must     | Complete history of all actions                     |
| FR-AD-002.19 | Priority flags (urgent/normal/low)                     | Must     | Visual indicator, sorting                           |
| FR-AD-002.22 | Export cases (CSV/Excel)                               | Must     | All case data exportable                            |

## Case Status Workflow

New → Assessment → Documents Collection → Document Preparation →  
 Application Submitted → Under Review (Authority) →  
 Decision Received → Post-Recognition → Closed

Additional statuses:

- On Hold (waiting for client action)
- Blocked (external issue)
- Cancelled
- Compensatory Measures Required

## FR-AD-003: Client Management (CRM)

| ID           | Requirement                               | Priority | Acceptance Criteria                                 |
|--------------|-------------------------------------------|----------|-----------------------------------------------------|
| FR-AD-003.1  | Client database (all registered users)    | Must     | Full list with profile info                         |
| FR-AD-003.2  | Lead management (from quiz, unregistered) | Must     | Leads separate from active clients                  |
| FR-AD-003.3  | Lead pipeline view                        | Must     | Lead → Contacted → Consultation → Proposal → Client |
| FR-AD-003.4  | Client profile view (admin)               | Must     | All info + cases + payments + communications        |
| FR-AD-003.5  | Communication history (unified)           | Must     | All emails, messages, calls in one timeline         |
| FR-AD-003.9  | Automated lead follow-up                  | Must     | Auto-email if no response in X days                 |
| FR-AD-003.16 | Import/export clients (CSV)               | Must     | For migration/backup                                |

| ID           | Requirement                            | Priority | Acceptance Criteria               |
|--------------|----------------------------------------|----------|-----------------------------------|
| FR-AD-003.17 | GDPR tools (export/delete client data) | Must     | One-click data export or deletion |

#### FR-AD-004: Document Review (Admin)

| ID          | Requirement                 | Priority | Acceptance Criteria                                   |
|-------------|-----------------------------|----------|-------------------------------------------------------|
| FR-AD-004.1 | Document review queue       | Must     | List of documents awaiting review                     |
| FR-AD-004.2 | In-browser document viewer  | Must     | View PDF/images without downloading                   |
| FR-AD-004.3 | Approve document            | Must     | One-click approve, status updates, client notified    |
| FR-AD-004.4 | Reject document with reason | Must     | Reason field (required), suggestions, client notified |
| FR-AD-004.5 | Add notes to document       | Must     | Internal notes about the document                     |

#### FR-AD-005: Messaging (Admin)

| ID          | Requirement                               | Priority | Acceptance Criteria                      |
|-------------|-------------------------------------------|----------|------------------------------------------|
| FR-AD-005.1 | Unified inbox (all client conversations)  | Must     | Single view of all messages              |
| FR-AD-005.2 | Filter by: unread, case, client, assigned | Must     | Quick filters                            |
| FR-AD-005.3 | Reply to client messages                  | Must     | From admin panel                         |
| FR-AD-005.4 | Canned responses / templates              | Must     | Quick insert pre-written messages        |
| FR-AD-005.5 | Internal notes within conversation        | Must     | Team can add notes without client seeing |
| FR-AD-005.6 | Assign conversation to team member        | Must     | Route to appropriate person              |
| FR-AD-005.8 | WhatsApp integration (receive/reply)      | Must     | WhatsApp messages appear in same inbox   |

#### FR-AD-006: Financial Management (Admin)

| ID          | Requirement                            | Priority | Acceptance Criteria                                    |
|-------------|----------------------------------------|----------|--------------------------------------------------------|
| FR-AD-006.1 | Revenue dashboard                      | Must     | Total revenue, by period, by service, by country       |
| FR-AD-006.2 | Invoice management                     | Must     | Create, view, send, mark as paid                       |
| FR-AD-006.3 | Invoice generation (Italian compliant) | Must     | Fattura format, sequential numbering, all legal fields |
| FR-AD-006.4 | Outstanding payments view              | Must     | Who owes what, days overdue                            |
| FR-AD-006.5 | Payment reminder sending               | Must     | Manual or automated reminders                          |
| FR-AD-006.6 | Record manual payment (bank transfer)  | Must     | When client pays via bonifico, admin records it        |
| FR-AD-006.7 | Refund processing                      | Must     | Initiate refund, track status                          |
| FR-AD-006.8 | Coupon/discount code management        | Must     | Create codes, set rules, track usage                   |
| FR-AD-006.9 | Financial reports (exportable)         | Must     | Monthly/yearly, by category, CSV/PDF                   |

#### FR-AD-007: Analytics & Reporting (Admin)

| ID          | Requirement                      | Priority | Acceptance Criteria                       |
|-------------|----------------------------------|----------|-------------------------------------------|
| FR-AD-007.1 | Key metrics overview (dashboard) | Must     | Leads, active cases, completed, revenue   |
| FR-AD-007.2 | Date range selector              | Must     | Today, week, month, quarter, year, custom |
| FR-AD-007.4 | Charts: revenue over time        | Must     | Line chart, configurable granularity      |
| FR-AD-007.5 | Charts: cases by country (pie)   | Must     | Distribution visualization                |



| ID           | Requirement                       | Priority | Acceptance Criteria                               |
|--------------|-----------------------------------|----------|---------------------------------------------------|
| FR-AD-007.6  | Charts: cases by profession (bar) | Must     | Distribution visualization                        |
| FR-AD-007.7  | Conversion funnel                 | Must     | Quiz → Lead → Consultation → Client               |
| FR-AD-007.8  | Average case duration             | Must     | By type, trend over time                          |
| FR-AD-007.12 | Alert system (anomalies)          | Must     | Cases overdue, payments late, leads not contacted |
| FR-AD-007.13 | Export reports (PDF, CSV)         | Must     | Download formatted reports                        |

## FR-AD-009: Automation Engine

| ID           | Requirement                   | Priority | Acceptance Criteria                                           |
|--------------|-------------------------------|----------|---------------------------------------------------------------|
| FR-AD-009.1  | Email template management     | Must     | Create/edit email templates with variables                    |
| FR-AD-009.2  | Triggered email sequences     | Must     | Set up auto-emails based on events                            |
| FR-AD-009.3  | Event triggers (configurable) | Must     | New lead, payment, document upload, status change, inactivity |
| FR-AD-009.4  | Delay between actions         | Must     | 'Wait 2 days, then send email'                                |
| FR-AD-009.7  | Task auto-creation            | Must     | Create tasks automatically on events                          |
| FR-AD-009.9  | Automation logs               | Must     | See what automation fired, when, for whom                     |
| FR-AD-009.10 | Enable/disable automations    | Must     | Toggle on/off without deleting                                |

### Core Automation Workflows

Automation 1: New Lead Welcome

Trigger: Quiz completed + email captured

Actions:

1. Create lead in CRM
2. Send welcome email (with quiz results)
3. Wait 1 day → Send follow-up email
4. Wait 3 days (if no purchase) → Send offer email
5. Create task for team: "Follow up with lead"

Automation 2: New Client Onboarding

Trigger: Payment received

Actions:

1. Create case
2. Assign to team member
3. Send welcome email (login details, next steps)
4. Send WhatsApp message ("Welcome! Here's what to do next")
5. Create tasks for team
6. Schedule check-in for day 3

Automation 3: Document Uploaded

Trigger: Client uploads document

Actions:

1. Add to review queue
2. Notify assigned team member
3. Create review task (due: 24 hours)

Automation 4: Case Status Change

Trigger: Admin changes case status

Actions:

1. Update timeline display for client
2. Notify client (email + WhatsApp + push)
3. If "Submitted" → send congratulations email
4. If "Decision Received" → send celebration email

Automation 5: Inactivity Nudge

Trigger: No client activity for 7 days (with pending actions)

Actions:

1. Send reminder email → Wait 3 days → WhatsApp
2. Wait 4 days → Create task: "Call client"

Automation 6: Case Completed

Trigger: Case status → Closed (successful)

Actions:

1. Send congratulations email
2. Wait 2 days → Send satisfaction survey
3. Wait 5 days → Send "Leave us a review" email
4. Wait 7 days → Send referral program invite

## 6. Non-Functional Requirements

### 6.1 Performance

| ID           | Requirement                         | Target                             | Measurement             |
|--------------|-------------------------------------|------------------------------------|-------------------------|
| NFR-PERF-001 | Page load time (initial)            | < 2 seconds                        | Lighthouse, WebPageTest |
| NFR-PERF-002 | Time to interactive                 | < 3 seconds                        | Lighthouse              |
| NFR-PERF-003 | API response time (95th percentile) | < 300ms                            | Server-side monitoring  |
| NFR-PERF-004 | Real-time message delivery          | < 1 second                         | WebSocket latency       |
| NFR-PERF-005 | File upload speed                   | Progressive, no timeout up to 20MB | Client-side testing     |
| NFR-PERF-006 | Search results return time          | < 500ms                            | Application monitoring  |
| NFR-PERF-007 | Concurrent users supported          | 1,000+                             | Load testing            |
| NFR-PERF-008 | Database query time (95th)          | < 100ms                            | Database monitoring     |
| NFR-PERF-009 | Core Web Vitals (LCP)               | < 2.5s                             | Google Search Console   |
| NFR-PERF-010 | Core Web Vitals (FID)               | < 100ms                            | Google Search Console   |
| NFR-PERF-011 | Core Web Vitals (CLS)               | < 0.1                              | Google Search Console   |

### 6.2 Availability & Reliability

| ID          | Requirement                    | Target                                 |
|-------------|--------------------------------|----------------------------------------|
| NFR-AVL-001 | Uptime                         | 99.9% (max 8.7 hrs downtime/year)      |
| NFR-AVL-002 | Planned maintenance windows    | Notify 48h in advance, max 2h duration |
| NFR-AVL-003 | Recovery Time Objective (RTO)  | < 1 hour                               |
| NFR-AVL-004 | Recovery Point Objective (RPO) | < 1 hour (max 1 hour data loss)        |
| NFR-AVL-005 | Error rate                     | < 0.1% of all requests                 |
| NFR-AVL-006 | Backup frequency               | Database: hourly. Files: daily         |
| NFR-AVL-007 | Backup retention               | 30 days                                |
| NFR-AVL-008 | Backup testing                 | Monthly restore test                   |

### 6.3 Scalability

| ID          | Requirement               | Target               |
|-------------|---------------------------|----------------------|
| NFR-SCL-001 | Registered users capacity | 100,000+             |
| NFR-SCL-002 | Active cases capacity     | 10,000+ simultaneous |
| NFR-SCL-003 | File storage capacity     | 10TB+ (expandable)   |

| ID          | Requirement            | Target                                   |
|-------------|------------------------|------------------------------------------|
| NFR-SCL-004 | Horizontal scaling     | Application servers auto-scale on demand |
| NFR-SCL-005 | Database scaling       | Read replicas, connection pooling        |
| NFR-SCL-006 | CDN for static content | Global CDN (Cloudflare/CloudFront)       |

## 6.4 Usability

| ID          | Requirement                | Target                                                  |
|-------------|----------------------------|---------------------------------------------------------|
| NFR-USB-001 | Mobile responsive          | All features usable on mobile (320px – 2560px)          |
| NFR-USB-002 | Accessibility              | WCAG 2.1 Level AA compliance                            |
| NFR-USB-003 | Browser support            | Chrome 90+, Firefox 90+, Safari 14+, Edge 90+           |
| NFR-USB-004 | Device support             | iOS 14+, Android 10+, desktop (Windows/Mac/Linux)       |
| NFR-USB-005 | Onboarding completion rate | 80%+ complete first action within 10 min of registering |
| NFR-USB-006 | Error messages             | Clear, actionable, in user's language                   |
| NFR-USB-007 | Form validation            | Real-time inline validation                             |

## 6.5 Maintainability

| ID          | Requirement             | Target                                               |
|-------------|-------------------------|------------------------------------------------------|
| NFR-MNT-001 | Code documentation      | All modules documented, API documented               |
| NFR-MNT-002 | Test coverage           | >70% unit test coverage, critical paths 100%         |
| NFR-MNT-003 | CI/CD pipeline          | Automated testing + deployment                       |
| NFR-MNT-004 | Environment setup       | Development, Staging, Production environments        |
| NFR-MNT-005 | Deployment frequency    | Multiple times per week (continuous delivery)        |
| NFR-MNT-006 | Rollback capability     | One-click rollback to previous version               |
| NFR-MNT-007 | Monitoring and alerting | Real-time alerts for errors, performance degradation |
| NFR-MNT-008 | Logging                 | Structured logging, searchable (ELK/Datadog)         |

## 7. User Stories

### 7.1 Epic: Visitor Discovery & Assessment

| ID     | User Story                                                                                                                                  | Key Acceptance Criteria                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| US-001 | As a visitor, I want to understand what Avanza does within 5 seconds of landing, so that I can determine if this service is relevant to me. | Value proposition clearly visible above the fold; in my language; clear CTA visible without scrolling                                              |
| US-002 | As a visitor, I want to take a quick assessment quiz, so that I can understand my specific situation and what I need.                       | Quiz takes less than 3 minutes; questions are clear; personalized results immediately; includes pathway, documents needed, timeline, cost estimate |
| US-003 | As a visitor, I want to see success stories from people like me, so that I can trust that Avanza can help me.                               | Testimonials from same country/profession if available; include name, photo, country, profession, outcome; verifiable                              |
| US-004 | As a visitor, I want to understand the costs clearly, so that I can decide if I can afford the service.                                     | Pricing displayed transparently; package comparison available; cost calculator; installment options mentioned                                      |
| US-005 | As a visitor, I want to read a detailed guide about my specific case, so that I can understand the process even if I decide to DIY.         | Country + profession specific page exists; step-by-step process explained; documents listed; realistic timeline; CTA to get professional help      |

### 7.2 Epic: Client Onboarding

| ID     | User Story                                                                                               | Key Acceptance Criteria                                                                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-010 | As a new client, I want to sign up quickly, so that I can start the process without friction.            | Sign up with email or Google (< 2 minutes); email verification sent immediately; onboarding wizard guides first steps                                             |
| US-011 | As a new client, I want to complete my profile, so that the team has all information needed for my case. | Profile completion indicator shows what's missing; fields relevant to case type; auto-save; guidance text for each field                                          |
| US-012 | As a new client, I want to book a consultation, so that I can discuss my case with an expert.            | See available slots in my timezone; book free or paid consultation; receive confirmation with calendar invite; can reschedule if needed                           |
| US-013 | As a new client, I want to choose and pay for a service package, so that my case can officially begin.   | See package options with clear feature comparison; pay by card, PayPal, or bank transfer; option to pay in installments; case created automatically after payment |

### 7.3 Epic: Document Management

| ID     | User Story                                                                                                             | Key Acceptance Criteria                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| US-020 | As a client, I want to see exactly which documents I need, so that I know what to collect without confusion.           | Personalized checklist for MY specific case; each item shows what it is, where to get it, any special requirements; status shown |
| US-021 | As a client, I want to upload documents easily from any device, so that I can make progress from my phone or computer. | Drag & drop on desktop; camera capture on mobile; support PDF, JPG, PNG (up to 20MB); upload progress shown                      |
| US-022 | As a client, I want to know when my documents are reviewed, so that I'm not left wondering about the status.           | Notification when document verified; notification when rejected with clear reason; can re-upload corrected document              |

| ID     | User Story                                                                                                  | Key Acceptance Criteria                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| US-024 | As a client, I want to receive reminders about missing documents, so that I don't forget and delay my case. | Reminder notification if document not uploaded within X days; escalating reminders; clear deadline date shown |

## 7.4 Epic: Case Tracking

| ID     | User Story                                                                                                                 | Key Acceptance Criteria                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| US-030 | As a client, I want to see the current status of my case at a glance, so that I feel in control and informed.              | Visual progress bar/timeline on dashboard; current step clearly highlighted; percentage completion shown; last update timestamp            |
| US-031 | As a client, I want to see a detailed timeline of my case, so that I can understand what has happened and what's coming.   | Step-by-step timeline with dates; completed steps shown; current step highlighted; future steps shown (greyed out)                         |
| US-032 | As a client, I want to know exactly what I need to do next, so that I can take action without waiting.                     | 'Your next steps' section clearly visible; action items have clear instructions; direct links to take action; deadline shown if applicable |
| US-033 | As a client, I want to receive notifications when my case progresses, so that I don't need to constantly check the portal. | Notification sent on every status change; via preferred channels; notification includes what changed, what it means, what to do next       |

## 7.5 Epic: Communication

| ID     | User Story                                                                                                             | Key Acceptance Criteria                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| US-040 | As a client, I want to message my case manager at any time, so that I can ask questions when they arise.               | Messaging accessible from dashboard and case view; can send text and attachments; response time expectation shown   |
| US-041 | As a client, I want to receive a response within 4 business hours, so that I don't feel ignored.                       | Team SLA: first response within 4 hours (business hours); after-hours auto-reply; escalation if SLA breached        |
| US-043 | As a client, I want to contact Avanza via WhatsApp, so that I can use a familiar tool without logging into the portal. | WhatsApp button available on website; messages appear in portal inbox too (synced); team can reply from admin panel |

## 7.6 Epic: Payments

| ID     | User Story                                                                                       | Key Acceptance Criteria                                                                                                     |
|--------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| US-050 | As a client, I want to pay easily and securely, so that I can start my service without friction. | Multiple payment options (card, PayPal, bank transfer); payment process takes < 2 minutes; secure (https, PCI compliant)    |
| US-051 | As a client, I want to pay in installments, so that the cost is manageable for me.               | Installment option clearly offered; see payment schedule; automatic charging on due dates; reminder before each installment |
| US-052 | As a client, I want to download my invoices, so that I can keep them for my records.             | Invoice generated for every payment; Italian-compliant format; downloadable PDF; also sent via email                        |

## 7.7 Epic: Admin Case Operations

| ID     | User Story                                                                                                                  | Key Acceptance Criteria                                                                                                                |
|--------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| US-060 | As a case manager, I want to see all my assigned cases at a glance, so that I can prioritize my work effectively.           | Dashboard shows my cases with status, priority, deadlines; sorted by urgency; alerts for overdue items                                 |
| US-061 | As a case manager, I want to update case status easily, so that clients stay informed of progress.                          | One-click status change; client automatically notified; internal note option; audit trail maintained                                   |
| US-062 | As a case manager, I want to add internal notes that clients can't see, so that I can track important internal information. | Clearly marked as 'internal' / 'team only'; timestamped with author; searchable; visible to all team members                           |
| US-063 | As a case manager, I want to review and verify client documents, so that only correct documents proceed to application.     | In-browser document preview; approve with one click; reject with mandatory reason + helpful suggestions; client automatically notified |

## 8. Information Architecture

### 8.1 Public Website Sitemap

```

/
■■■ / (Homepage)
■■■ /services
■   ■■■ /services/consultation
■   ■■■ /services/document-preparation
■   ■■■ /services/application-submission
■   ■■■ /services/full-service
■   ■■■ /services/business-registration
■■■ /countries
■   ■■■ /countries/romania
■   ■■■ /countries/albania
■   ■■■ /countries/ukraine
■   ■■■ /countries/morocco
■   ■■■ /countries/india
■   ■■■ /countries/[country-slug]
■■■ /professions
■   ■■■ /professions/electrician
■   ■■■ /professions/nurse
■   ■■■ /professions/engineer
■   ■■■ /professions/[profession-slug]
■■■ /[country]/[profession] (combination pages)
■   ■■■ /romania/electrician
■   ■■■ /albania/engineer
■■■ /quiz           ■■■ /calculator   ■■■ /booking
■■■ /blog           ■■■ /resources   ■■■ /about
■■■ /contact        ■■■ /pricing     ■■■ /faq
■■■ /sitemap.xml

```

### 8.2 Client Portal Structure

```

/portal
■■■ /portal/dashboard
■■■ /portal/cases
■   ■■■ /portal/cases/[case-id]
■   ■   ■■■ /portal/cases/[case-id]/timeline
■   ■   ■■■ /portal/cases/[case-id]/documents
■   ■   ■■■ /portal/cases/[case-id]/messages
■   ■   ■■■ /portal/cases/[case-id]/payments
■   ■■■ /portal/cases/new
■■■ /portal/documents
■■■ /portal/messages
■■■ /portal/appointments
■■■ /portal/payments
■   ■■■ /portal/payments/history
■   ■■■ /portal/payments/invoices
■■■ /portal/notifications
■■■ /portal/profile
    ■■■ /portal/profile/edit
    ■■■ /portal/profile/security
    ■■■ /portal/profile/preferences

```

### 8.3 Admin Dashboard Structure

```

/admin
■■■ /admin/dashboard
■■■ /admin/cases      (pipeline, list, calendar, [case-id])
■■■ /admin/clients    (all, leads, [client-id])

```



```
■■■ /admin/messages (inbox, [conversation-id], templates)
■■■ /admin/documents (review-queue, templates)
■■■ /admin/finance (overview, invoices, payments, coupons)
■■■ /admin/appointments
■■■ /admin/analytics (overview, revenue, cases, leads, team)
■■■ /admin/content (blog, pages, countries, professions, faq,
■ testimonials, translations)
■■■ /admin/automation (workflows, email-templates, logs)
■■■ /admin/team (members, roles, activity-log)
■■■ /admin/settings (general, pricing, notifications,
integrations, security)
```

# 9. Data Model

## 9.1 Entity Relationship Diagram (Simplified)

users (1) █████ (many) cases  
users (1) █████ (many) documents  
users (1) █████ (many) payments  
users (1) █████ (many) appointments  
users (1) █████ (many) notifications  
users (1) █████ (many) messages  
users (1) █████ (many) quiz\_submissions

cases (1) █████ (many) documents  
cases (1) █████ (many) messages (via conversations)  
cases (1) █████ (many) payments  
cases (1) █████ (many) tasks  
cases (1) █████ (many) activity\_logs  
cases (1) █████ (many) appointments

conversations (1) ██ (many) messages  
conversations (1) ██ (1) case (optional)  
conversations (1) ██ (1) client

countries (1) █████ (many) country\_profession\_guides  
professions (1) ██ (many) country\_profession\_guides

## 9.2 Core Schema

### Users Table

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email VARCHAR(255) UNIQUE NOT NULL,  
  password_hash VARCHAR(255),  
  role ENUM('client','admin','case_manager','support','content','finance') DEFAULT 'client',  
  first_name VARCHAR(100), last_name VARCHAR(100),  
  phone VARCHAR(20), date_of_birth DATE,  
  nationality VARCHAR(100), country_of_residence VARCHAR(100),  
  city VARCHAR(100), codice_fiscale VARCHAR(16),  
  country_of_qualification VARCHAR(100),  
  qualification_type ENUM('vocational','diploma','bachelor','master','phd','professional_license'),  
  profession VARCHAR(200), years_of_experience INTEGER,  
  preferred_language VARCHAR(10) DEFAULT 'it',  
  communication_preference JSONB,  
  email_verified BOOLEAN DEFAULT FALSE,  
  two_factor_enabled BOOLEAN DEFAULT FALSE,  
  marketing_consent BOOLEAN DEFAULT FALSE,  
  referral_code VARCHAR(20) UNIQUE,  
  referred_by UUID REFERENCES users(id),  
  created_at TIMESTAMP DEFAULT NOW(),  
  deleted_at TIMESTAMP -- soft delete  
);
```

### Cases Table

```
CREATE TABLE cases (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  case_number VARCHAR(20) UNIQUE NOT NULL, -- e.g. "AV-2024-0142"  
  user_id UUID NOT NULL REFERENCES users(id),  
  assigned_to UUID REFERENCES users(id),  
  title VARCHAR(300),
```

```

type ENUM('professional_recognition_eu','professional_recognition_non_eu',
          'dichiarazione_di_valore','equipollenza','cimea_attestation',
          'professional_order','business_registration') NOT NULL,
country_of_qualification VARCHAR(100) NOT NULL,
profession VARCHAR(200) NOT NULL,
goal ENUM('employment','self_employment','study','other') NOT NULL,
status ENUM('new','assessment','documents_collection','document_preparation',
            'application_submitted','under_review','compensatory_measures',
            'decision_received','post_recognition','closed_successful',
            'closed_cancelled','on_hold','blocked') DEFAULT 'new',
progress_percentage INTEGER DEFAULT 0,
priority ENUM('low','normal','high','urgent') DEFAULT 'normal',
package_type ENUM('starter','standard','premium','custom'),
package_price DECIMAL(10,2), total_paid DECIMAL(10,2) DEFAULT 0,
created_at TIMESTAMP DEFAULT NOW()
);

```

## Documents Table

```

CREATE TABLE documents (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  case_id UUID NOT NULL REFERENCES cases(id),
  user_id UUID NOT NULL REFERENCES users(id),
  document_type ENUM('diploma','transcript','apostille','sworn_translation',
                    'dichiarazione_di_valore','experience_certificate',
                    'identity_document','codice_fiscale_doc','residence_permit',
                    'professional_attestation','application_form','cover_letter',
                    'power_of_attorney','other') NOT NULL,
  file_name VARCHAR(300) NOT NULL,
  file_url TEXT NOT NULL,
  file_size INTEGER, mime_type VARCHAR(100),
  file_hash VARCHAR(64), -- SHA-256 for integrity
  status ENUM('uploaded','under_review','verified','rejected','expired') DEFAULT 'uploaded',
  rejection_reason TEXT, rejection_instructions TEXT,
  is_required BOOLEAN DEFAULT TRUE,
  deadline DATE, version INTEGER DEFAULT 1,
  uploaded_at TIMESTAMP DEFAULT NOW()
);

```

## Messages & Conversations

```

CREATE TABLE conversations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  case_id UUID REFERENCES cases(id),
  client_id UUID NOT NULL REFERENCES users(id),
  assigned_to UUID REFERENCES users(id),
  status ENUM('open','waiting_client','waiting_team','closed') DEFAULT 'open',
  last_message_at TIMESTAMP,
  unread_count_client INTEGER DEFAULT 0,
  unread_count_team INTEGER DEFAULT 0,
  created_at TIMESTAMP DEFAULT NOW()
);

CREATE TABLE messages (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  conversation_id UUID NOT NULL REFERENCES conversations(id),
  sender_id UUID NOT NULL REFERENCES users(id),
  body TEXT NOT NULL,
  message_type ENUM('text','file','image','audio','system') DEFAULT 'text',
  is_internal BOOLEAN DEFAULT FALSE, -- team-only note
  is_read BOOLEAN DEFAULT FALSE,
  source ENUM('portal','whatsapp','email','sms','system') DEFAULT 'portal',
  reply_to_id UUID REFERENCES messages(id),
  created_at TIMESTAMP DEFAULT NOW()
);

```

```
);
```

## Payments & Invoices

```
CREATE TABLE payments (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL REFERENCES users(id),  
  case_id UUID REFERENCES cases(id),  
  amount DECIMAL(10,2) NOT NULL, currency VARCHAR(3) DEFAULT 'EUR',  
  payment_type ENUM('package','consultation','addon','installment') NOT NULL,  
  payment_method ENUM('stripe_card','paypal','bank_transfer','other') NOT NULL,  
  status ENUM('pending','processing','completed','failed','refunded') DEFAULT 'pending',  
  stripe_payment_id VARCHAR(255), paypal_order_id VARCHAR(255),  
  bank_transfer_reference VARCHAR(100),  
  paid_at TIMESTAMP, due_date DATE, created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE invoices (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  invoice_number VARCHAR(20) UNIQUE NOT NULL, -- "INV-2024-0089"  
  user_id UUID NOT NULL REFERENCES users(id),  
  amount DECIMAL(10,2) NOT NULL, tax_amount DECIMAL(10,2) DEFAULT 0,  
  total_amount DECIMAL(10,2) NOT NULL, currency VARCHAR(3) DEFAULT 'EUR',  
  status ENUM('draft','sent','paid','overdue','cancelled') DEFAULT 'draft',  
  issued_date DATE NOT NULL, due_date DATE NOT NULL, paid_date DATE,  
  pdf_url TEXT, sdi_submitted BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

## 10. API Specifications

### 10.1 API Design Principles

- RESTful API architecture
- JSON request/response format
- JWT-based authentication
- API versioning: /api/v1/...
- Rate limiting: 100 req/min for authenticated, 30 for anonymous
- Pagination: cursor-based for lists
- Error format: {"error": {"code": "...", "message": "...", "details": [...]}}
- CORS configured for allowed origins
- Request/response logging (sanitized)

### 10.2 Authentication Endpoints

|      |                              |                               |
|------|------------------------------|-------------------------------|
| POST | /api/v1/auth/register        | - Create new account          |
| POST | /api/v1/auth/login           | - Login (returns JWT)         |
| POST | /api/v1/auth/logout          | - Invalidate token            |
| POST | /api/v1/auth/refresh         | - Refresh JWT token           |
| POST | /api/v1/auth/forgot-password | - Send reset email            |
| POST | /api/v1/auth/reset-password  | - Set new password with token |
| POST | /api/v1/auth/verify-email    | - Verify email with token     |
| POST | /api/v1/auth/social/google   | - Google OAuth login          |
| POST | /api/v1/auth/social/facebook | - Facebook OAuth login        |
| POST | /api/v1/auth/2fa/enable      | - Enable 2FA                  |
| POST | /api/v1/auth/2fa/verify      | - Verify 2FA code             |

### 10.3 User Endpoints

|               |                              |                              |
|---------------|------------------------------|------------------------------|
| GET           | /api/v1/users/me             | - Get current user profile   |
| PUT           | /api/v1/users/me             | - Update profile             |
| PUT           | /api/v1/users/me/password    | - Change password            |
| PUT           | /api/v1/users/me/preferences | - Update preferences         |
| DELETE        | /api/v1/users/me             | - Delete account (GDPR)      |
| GET           | /api/v1/users/me/export      | - Export user data (GDPR)    |
| # Admin only: |                              |                              |
| GET           | /api/v1/admin/users          | - List all users (paginated) |
| GET           | /api/v1/admin/users/:id      | - Get specific user          |
| PUT           | /api/v1/admin/users/:id      | - Update user (admin)        |

### 10.4 Case Endpoints

|           |                                |                                          |
|-----------|--------------------------------|------------------------------------------|
| # Client: |                                |                                          |
| GET       | /api/v1/cases                  | - List my cases                          |
| GET       | /api/v1/cases/:id              | - Get case details                       |
| GET       | /api/v1/cases/:id/timeline     | - Get case timeline/activity             |
| GET       | /api/v1/cases/:id/checklist    | - Get document checklist                 |
| # Admin:  |                                |                                          |
| GET       | /api/v1/admin/cases            | - List all cases (paginated, filterable) |
| POST      | /api/v1/admin/cases            | - Create new case                        |
| PUT       | /api/v1/admin/cases/:id        | - Update case                            |
| PUT       | /api/v1/admin/cases/:id/status | - Change status                          |
| PUT       | /api/v1/admin/cases/:id/assign | - Assign to team member                  |
| POST      | /api/v1/admin/cases/:id/notes  | - Add internal note                      |
| GET       | /api/v1/admin/cases/pipeline   | - Get pipeline view data                 |
| GET       | /api/v1/admin/cases/export     | - Export cases CSV                       |

## 10.5 Document Endpoints

|           |                                       |                                  |
|-----------|---------------------------------------|----------------------------------|
| # Client: |                                       |                                  |
| GET       | /api/v1/cases/:caseId/documents       | - List documents                 |
| POST      | /api/v1/cases/:caseId/documents       | - Upload document                |
| GET       | /api/v1/documents/:id/download        | - Download file                  |
| PUT       | /api/v1/documents/:id                 | - Replace document (new version) |
| # Admin:  |                                       |                                  |
| GET       | /api/v1/admin/documents/review-queue  | - Awaiting review                |
| PUT       | /api/v1/admin/documents/:id/verify    | - Approve document               |
| PUT       | /api/v1/admin/documents/:id/reject    | - Reject document (with reason)  |
| POST      | /api/v1/admin/cases/:caseId/documents | - Upload on behalf of client     |

## 10.6 Messaging Endpoints

|              |                                          |                            |
|--------------|------------------------------------------|----------------------------|
| # Client:    |                                          |                            |
| GET          | /api/v1/conversations                    | - List my conversations    |
| GET          | /api/v1/conversations/:id/messages       | - Get messages (paginated) |
| POST         | /api/v1/conversations/:id/messages       | - Send message             |
| PUT          | /api/v1/conversations/:id/read           | - Mark as read             |
| # Admin:     |                                          |                            |
| GET          | /api/v1/admin/conversations              | - List all (filterable)    |
| POST         | /api/v1/admin/conversations/:id/messages | - Reply                    |
| POST         | /api/v1/admin/conversations/:id/notes    | - Add internal note        |
| PUT          | /api/v1/admin/conversations/:id/assign   | - Assign conversation      |
| # WebSocket: |                                          |                            |
| WS           | /ws/conversations/:id                    | - Real-time messages       |

## 10.7 Payment Endpoints

|             |                                   |                                |
|-------------|-----------------------------------|--------------------------------|
| # Client:   |                                   |                                |
| GET         | /api/v1/payments                  | - My payment history           |
| POST        | /api/v1/payments/create-intent    | - Create Stripe payment intent |
| POST        | /api/v1/payments/confirm          | - Confirm payment              |
| GET         | /api/v1/invoices/:id/download     | - Download invoice PDF         |
| POST        | /api/v1/payments/apply-coupon     | - Validate and apply coupon    |
| # Admin:    |                                   |                                |
| GET         | /api/v1/admin/payments            | - All payments                 |
| POST        | /api/v1/admin/payments/manual     | - Record manual payment        |
| POST        | /api/v1/admin/payments/:id/refund | - Process refund               |
| POST        | /api/v1/admin/invoices            | - Create invoice               |
| # Webhooks: |                                   |                                |
| POST        | /api/v1/webhooks/stripe           | - Stripe webhook handler       |
| POST        | /api/v1/webhooks/paypal           | - PayPal webhook handler       |

## 10.8–10.13 Additional Endpoints

|                  |                                      |                            |
|------------------|--------------------------------------|----------------------------|
| # Appointments:  |                                      |                            |
| GET              | /api/v1/appointments/available-slots | - Get available time slots |
| POST             | /api/v1/appointments                 | - Book appointment         |
| PUT              | /api/v1/appointments/:id/reschedule  | - Reschedule               |
| PUT              | /api/v1/appointments/:id/cancel      | - Cancel                   |
| # Notifications: |                                      |                            |
| GET              | /api/v1/notifications                | - My notifications         |
| GET              | /api/v1/notifications/unread-count   | - Get unread count         |
| PUT              | /api/v1/notifications/read-all       | - Mark all as read         |
| # Quiz & Leads:  |                                      |                            |
| POST             | /api/v1/quiz/submit                  | - Submit quiz, get results |

```
GET    /api/v1/admin/leads                - All leads
PUT    /api/v1/admin/leads/:id/convert  - Convert lead to client

# Analytics (Admin):
GET    /api/v1/admin/analytics/overview - Dashboard KPIs
GET    /api/v1/admin/analytics/revenue  - Revenue data
GET    /api/v1/admin/analytics/funnel   - Conversion funnel

# Automation (Admin):
GET    /api/v1/admin/automation/workflows - List workflows
PUT    /api/v1/admin/automation/workflows/:id/toggle - Enable/disable
GET    /api/v1/admin/automation/logs      - Execution logs
```

# 11. Third-Party Integrations

## 11.1 Required Integrations (Phase 1)

| Service                | Purpose               | Priority | API/SDK                             |
|------------------------|-----------------------|----------|-------------------------------------|
| Stripe                 | Payment processing    | Must     | Stripe.js + Server SDK              |
| PayPal                 | Alternative payment   | Must     | PayPal REST API                     |
| Google OAuth           | Social login          | Must     | Google Identity Services            |
| Google Calendar        | Appointment sync      | Must     | Google Calendar API v3              |
| Google Meet            | Video meetings        | Must     | Google Meet API                     |
| SendGrid/Postmark      | Transactional email   | Must     | REST API                            |
| Twilio                 | WhatsApp + SMS        | Must     | Twilio REST API + WhatsApp Business |
| AWS S3 / Cloudflare R2 | File storage          | Must     | S3 API                              |
| Cloudflare             | CDN + DDoS protection | Must     | DNS + proxy                         |
| Google Analytics 4     | Website analytics     | Must     | GA4 Measurement Protocol            |

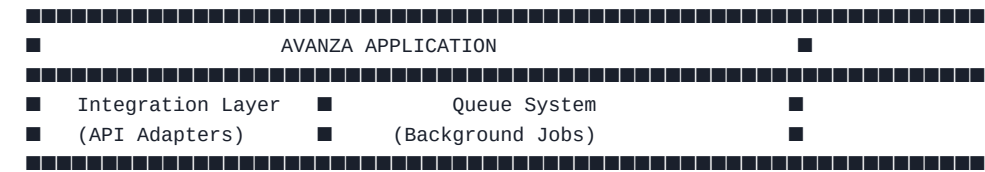
## 11.2 Phase 2 Integrations

| Service          | Purpose                         | Priority |
|------------------|---------------------------------|----------|
| Fatture in Cloud | Italian electronic invoicing    | Should   |
| DeepL            | Translation assistance          | Should   |
| Zapier/Make      | External automation connections | Should   |
| Facebook Pixel   | Ad tracking                     | Should   |
| Google Ads       | Conversion tracking             | Should   |
| ClamAV           | Virus scanning for uploads      | Must     |
| Firebase         | Push notifications (web/mobile) | Should   |

## 11.3 Phase 3 Integrations

| Service        | Purpose                    | Priority |
|----------------|----------------------------|----------|
| OpenAI         | AI document analysis       | Could    |
| Zoom           | Alternative video meetings | Could    |
| Intercom/Crisp | Advanced live chat         | Could    |
| Plausible      | Privacy-friendly analytics | Could    |

## 11.4 Integration Architecture





Page 33